SYSTEM ARCHITECTURE & DESIGN DOCUMENT

Technology Hub Inventory Management System

PLATFORM	APPLICATION ID	SCHEMA	DATABASE
Oracle APEX 26.2.0	156569	WKSP_DOULGAS	Oracle 19c Enterprise

1. ARCHITECTURE OVERVIEW

The system is a monolithic Oracle APEX application utilizing the Universal Theme (Theme 42). It operates on a single database schema with client-side rendering for Interactive Reports (IR) and server-side PL/SQL processes for secure data manipulation.

- **Authentication:** Custom Authentication Scheme (`custom_authenticate`) querying the `APP_USERS` table. Passwords are secured using SHA-256 hashing.
- **Authorization:** Role-Based Access Control (RBAC) utilizing "Must Be Admin" and "Must Be Manager or Admin" PL/SQL function bodies applied at the page level.
- **Data Model:** Relational schema comprising core domain entities (`PRODUCTS` , `CATEGORIES` , `SUPPLIERS` , `CUSTOMERS` , `SALES` , `SALE_ITEMS` , `INVENTORY_TRANSACTIONS` , `APP_USERS` , `AUDIT_LOG`).

2. DATABASE SCHEMA & RELATIONSHIPS

- **`PRODUCTS`** : Core inventory table. Linked to `CATEGORIES` and `SUPPLIERS` . Tracks `SKU` , `QUANTITY` , `REORDER_LEVEL` , and pricing in UGX.
- **`SALES`** : Header table for transaction records. Linked directly to `CUSTOMERS` .
- **`SALE_ITEMS`** : Line-item details linking `SALES` to `PRODUCTS` . Tracks `QUANTITY` and `UNIT_PRICE` .
- **`INVENTORY_TRANSACTIONS`** : Audit trail for stock adjustments (`STOCK_IN` , `STOCK_OUT` , `ADJUSTMENT`).
- **`APP_USERS`** : Application user directory containing `USERNAME` , `PASSWORD` (SHA-256 Hash), `ROLE` , and `STATUS` .

3. APPLICATION PAGE MAP (26 PAGES)

PAGE ID	PAGE NAME	TYPE	SECURITY	DESCRIPTION
1	Home	Standard	None	Landing page with navigation links.
2, 3	Categories	IR / Form	Manager / Admin	CRUD for product categories.
4, 5	Suppliers	IR / Form	Manager / Admin	CRUD for supplier details.
6, 7	Products	IR / Form	Manager / Admin	CRUD for products. Includes SKU uniqueness validation.
8, 9	Customers	IR / Form	None	CRUD for customer details.
10, 11	Sales	IR / Form	None	Header-level sales records.
12, 13	Sale Items	IR / Form	None	Line-item details. Subtotal calculated server-side.
14, 15	Inventory Trans.	IR / Form	Manager / Admin	Manual stock transaction logging.
16, 17	App Users	IR / Form	Admin	User management (Create/Edit roles).

PAGE ID	PAGE NAME	TYPE	SECURITY	DESCRIPTION
18, 19	Audit Log	IR / Read-Only	Admin	System audit trail.
20	Dashboard	Dashboard	None	9 KPI Cards, Sales Overview Chart, Low Stock Report.
21	Inventory Reports	IR	Manager / Admin	Stock valuation and low-stock status.
23	Low Stock Report	IR	Manager / Admin	Filtered view of items below reorder level.
25	Sales Reports	IR	Manager / Admin	Historical sales data.
26	Customer Receipt	SQL Report	None	Printable receipt interface.

4. BUSINESS LOGIC & AUTOMATION

- **Inventory Updates:** When a record is created in `INVENTORY_TRANSACTIONS` (Page 15), an `ON_SUBMIT_BEFORE_COMPUTATION` PL/SQL process updates `PRODUCTS.QUANTITY` based on the `STOCK_IN`, `STOCK_OUT`, or `ADJUSTMENT` type.
- **Sales Processing:** When a `SALE_ITEMS` record is created (Page 13), PL/SQL processes automatically:
  1. Calculate the Subtotal securely on the server.
  2. Deduct the quantity from `PRODUCTS.QUANTITY`.
  3. Recalculate the parent `SALES.TOTAL_AMOUNT` by summing all `SALE_ITEMS.SUBTOTAL`.